



MEMO

TO: Honorable Mayor Patrick J. Collins, City of Cheyenne
Members of the City Council, City of Cheyenne

FROM: Sheree Brgoch, Associate Staff Engineer

DATE: May 19, 2026

SUBJECT: Contraction Modification Number 2 to Professional Service Agreement between the City of Cheyenne and HDR Inc, for the design of 18th Street Reconstruction Project.

Please find the attached Contract Modification between the City of Cheyenne and HDR Inc. for engineering services related to the design of the 18th Street Reconstruction Project.

The new contract modification reflects a scope change that was needed to alter the outfall location of the proposed storm sewer interceptor being designed in the 18th street corridor.

This contract modification also extends the contract deadline to December 31, 2026. This extension is needed due to the additional findings while in the design process.

The additional costs associated with this work will be \$313,197.00. Funding for this project will come from the Drainage GL 021-26-2618-42-43601 and Street Improvements GL 021-26-2616-42-43498 accounts. Your approval of this contract modification is respectfully requested.

CITY OF CHEYENNE
CONTRACT MODIFICATION

APPROVED as to form only:
By David Hopkinson at 11:23 am, May 05 2026

Contract Modification: #2
Project: 18th Street Reconstruction Project
Project Number: Contract No. 8199
Contractor: HDR Engineering, Inc.
Date Prepared: April 20, 2026

The above referenced contract is amended or modified as set forth in the following and these amendments and modifications of the plans and specifications are agreed to by the parties. All other aspects of the original agreement remain unchanged and in full force and effect.

I. Description of Changes:

This contract modification reflects a substantial scope change that was needed to alter the outfall location of the proposed storm sewer interceptor being designed in the 18th street corridor. The proposed interceptor will include the design and routing of a storm sewer interceptor south down Bent Ave to 15th street and will be designed to connect to the existing overflow structure located immediately north of UPRR property on an already developed wetlands and stormwater storage pond. The professional design services are described in further detail in Exhibit A.

II. New Contract Items:

Additional professional services described above, and in detail in Exhibit A.

III. Schedule:

Additional time for contract completion date is December 31, 2026.

IV. Summary:

Original Contract Amount	\$1,100,562.00
Previous Modifications	\$28,046.00
Contract Amount Prior to this Modification	\$1,128,608.00
This Modification	\$313,197.00
New Contract Amount	\$1,441,805.00

V. GL Account Information:

If requesting a modification to the contract price, please list GL account # to cover the balance of this modification (if using more than (1) GL account, list the dollar amount for each respective account):

GL# 021-26-2618-42-43601 Amount: \$156,598.50
GL# 021-26-2616-42-43498 Amount: \$156,598.50

VI. Approvals:

City Project Manager: _____ Date: _____

Consultant/Contractor: _____ Date: _____

City of Cheyenne DATE: _____

Mayor Patrick Collins /
/ATTEST: City Clerk Kylie Soden

Scope of Services – Amendment No.2

March 16th, 2026

BACKGROUND

Stemming from a detailed Hydraulic & Hydrologic assessment discussion with the City, HDR, CivilWorx, and GLM Design it was determined that a substantial scope change was needed to alter the outfall location of the proposed storm sewer interceptor being designed in the 18th Street Corridor. The new proposed interceptor routing will not include the design of an outfall structure. Instead, it will include the design and routing of storm sewer interceptor south down Bent Avenue to 15th Street and will be designed to connect to the existing overflow structure located immediately north of UPRR property on an already developed wetlands and stormwater storage pond. At the request of the City a new detention pond will be designed, in the future by others. Finally, this scope modification also includes the removal of any remaining scoped work from Warren Avenue to House Avenue as it will be included in a future, separate, project.

HDR has developed the below scope of services to accommodate the described additional work. This scope of services has been discussed with City Staff and represents HDR's evaluation and interpretation of the level of effort required and the assumptions necessary to complete this additional work.

TASK SERIES 1.0 – General Conditions

Tasks 1.1 through 1.4

- Task Series 1.0 General Conditions has been developed to capture overall project administration, project management, and quality control costs throughout the duration of the project
 - A minimal amount of additional Project Management time has been included to account for the rescoping and development of the new storm sewer outfall.
- Two additional months of Project Management time have been added to non-storm water related design components.
 - This includes an additional design review meeting and budget update per month of additional time.

TASK SERIES 2.0 – Preliminary Design Phase

Task 2.1 – Surveying & Base Mapping

- HDR will perform additional data collection and field survey for Bent Avenue from 15th Street to 18th Street AND a 100' wide path from the 15th Street and Bent Ave. Intersection to Ames Avenue as follows:
 - Establish Horizontal and Vertical Control
 - Topographic Survey of roadway centerline, shoulders, C&G, and other hard surfacing features
 - Survey surface features such as utilities (public and private), fences, mailboxes, signs, etc.
 - Survey existing utility inverts including photograph conditions and measure sizes

- Property Corners will be recorded based on number of corners able to be found on-site during the duration of the data collection and field survey task. Corners will be found using a mag-locator and recorded where discovered
- Private Utility Locations will be recorded based on One-Call of Wyoming's provided field markings
 - Additional coordination with identified utility owners is anticipated in order to verify what has been field located
 - Coordination will include post survey mapping review to evaluate significant public and private utilities
- Subsurface utility exploration is anticipated and will be performed in conjunction with data collection efforts
- It is assumed that no more than 10 / EA exploratory locations will be required. This has been estimated by reviewing available City of Cheyenne (BOPU) mapping and reviewing 811 field locates provided prior to completion of this Scope of Work.
 - No SUE locates are anticipated west of 15th street near UPRR property
- Assumptions:
 - Right-of-Way acquisition is NOT anticipated, or will be completed by others, and as such legal exhibits are not anticipated

Task 2.2 – Project Kick-off & Conceptual Design

- HDR will attend a re-scoping meeting with the City to finalize storm interceptor routing into the new project area
- 2 - Field visit / on-site investigation – Two people will review the site with City Staff
- Establish Project Limits and estimated project impact areas for any future cultural resources coordination items

Task 2.3 – Geotechnical Investigation

- HDR has partnered with Terracon to perform additional Geotechnical Investigation Services
 - The scope of services from Terracon has been attached for reference
 - Refer to attached Terracon Proposal for additional information

Task 2.4 – Roadway and Intersection Geometrics

- Horizontal Alignment layout will be completed with City input. Adjustments to the existing alignment are not anticipated
 - Vertical Alignment will match existing with minimal change to street grade(s)
 - Based on City feedback, no reconstruction/reconfiguration of existing street(s) is anticipated. Repair/replacement of roadway will be designed only for the storm sewer interceptor itself.
 - We assume no c&g, sidewalks, valley gutters, fillets, etc. Will need to be redesigned or replaced.

Task 2.5 – Preliminary Plans (Submittal)

- HDR will provide preliminary design plan and profile sheets of the additional project area infrastructure for review by City
 - Full quantity take-offs will not be completed during this stage of review

- Cross Sections will not be completed for this stage of review
- City review time has been estimated to be completed and returned in 5 working days in order to maintain proposed project schedule

Task 2.8 – Baseline Hydrologic & Hydraulic Analysis & Drainage Impact Study

- HDR will expand the drainage design and analysis from the existing conditions model that has been established for the Capitol Basin area to develop improvements and sizing for new infrastructure.
 - Review existing model and understand model logic and linkages
- Model Review and Updates
 - HDR will apply industry standard practices in modelling to perform a comprehensive review and update of the existing model
 - Recommendations will be made for model improvements and implementation
- HDR will incorporate the additional project area into the Drainage Impact Study for the City's review. As stated in the original scope of work this study will comply with Section 3.2 of the UDC and will assess the following:
 - Upstream and downstream impacts
 - Compliance with drainage criteria for storm sewer design
 - Preliminary Drainage Plan will be provided at this stage
 - Define minimum project goals
 - Define issues/opportunities/fatal flaws that drive the concept plan (project goals, utility conflicts, grade, ROW/easements, etc).
 - Review the Existing model
 - Update the existing model for recent projects
 - Develop concept design alternative(s) based on input from the City and the results of the issues/opportunities/fatal flaw analysis

Task 2.9 – Utility Design & Coordination (BOPU)

- HDR will evaluate and identify any conflicts with existing water and sanitary sewer system and proposed extension of the storm water interceptor.
 - It is assumed that minimal relocation of BOPU utilities will be required.
 - This includes 1 sanitary sewer manhole adjustment and 1 waterline lowering
 - Replacement of the BOPU waterline extending to the west from 15th Street is not included in this amendment as we have been informed that it can be abandoned by the City & BOPU.

Task 2.10 – 60% Plans

- HDR will incorporate additional project area and design improvements into the 60% Plans as originally scoped. Generally, this includes plan and profile sheets with limited quantities for review by City
 - Full quantity take-offs will not be completed during this stage of review but shall be furthered from the Preliminary Plans Submittal
 - Cross Sections will not be completed for this stage of review
- Project total quantities will be estimated to prepare a preliminary cost estimate
 - HDR will follow the American Association of Cost Estimators guidelines during this effort
- Bid items will be based on Technical Specifications (City of Cheyenne & WYDOT)
- Preliminary project cost estimate will be provided at this design review
 - City review time has been estimated to be completed and returned in 5 working days in order to maintain proposed project schedule

- It is assumed that intersection geometry and horizontal location of proposed new utilities will be largely finalized for this plan review. Minor adjustments to accommodate ROW limitations and/or utility conflicts will be established in the 90% Plans submittal

TASK SERIES 3.0 – Final Design Phase

Task 3.2 a&b – Detailed Underground Utility Design (Water, Sewer, Storm Sewer)

- For the additional work area HDR will continue to evaluate existing utilities and work to mitigate potential conflicts with proposed improvements.
 - Public utilities include water, sanitary sewer, and storm sewer
 - No adjustments/replacement of existing storm sewer is anticipated and there is currently little to no storm infrastructure in Bent Avenue.
- HDR will include the additional project area in the final conditions model for the 18th Street and Bent Avenue area updates along with address and remaining comments on the Drainage Impact Study.
- HDR has included structural design time to include three (3) structurally designed junction (and/or flow separator) boxes to transition from large diameter storm sewer interceptor to multiple smaller diameter pipe(s) to traverse the railroad spur line.
 - The junction boxes may be used to change direction from south to west at the intersection of 15th and Bent and/or to separate flows for what is assumed to be a railroad preferred crossing type/style.
 - Junction box design also includes design time to create a connection to the existing overflow box structure in the existing wetlands/detention area.

Task 3.3 – H&H Drainage Model Updates

- HDR will refine storm sewer design and model to include the additional project area throughout the progression of final design
 - This includes the final model deliverable reflecting the final design plans

Task 3.7 – 90% Design Plans

- 90% Design Plans will be submitted to include additional project area and designed improvements described above.

Task 3.9 – Final Design (Plans, Specifications, Estimate)

- HDR will include additional project area with Final Design plans and Specifications for bidding in accordance with original scope of work.
 - It is assumed that the additional project area will add no more than 8 additional plan sheets.

Subconsultants

- HDR has partnered with Badger Daylighting and Terracon to provide additional services as described herein. Please refer to proposal(s) from teaming partners for detailed scope of services

Assumptions

- No permitting has been included as part of the storm sewer interceptor design.
 - Future 404, 401, UPRR, and SEO permits will be obtained by the City of Cheyenne prior to construction.
- City of Cheyenne is responsible for all permitting and approvals from the BNSF and/or the UPRR for surveying and access.
- No revision to FEMA flood mapping is included as part of this amendment.
 - A CLOMR/LOMR will be completed in the future by others.

TASK SERIES 5.0 – FEMA Benefit-Cost Analysis

HDR will prepare a benefit-cost analysis to support a future FEMA grant application for the stormwater portion of the project. The benefit-cost analysis will be prepared by HDR in accordance with FEMA guidelines.

Task 5.1 – Data Gathering

The completion of the benefit-cost analysis will require technical data to support the estimates of annual damages and annual benefits. The technical data required is as follows:

Data Required for Estimating Annual Damages with Project

- The flood events & depth raster for four recurrence intervals (corresponding to those evaluated in Task 2.8).
- Estimated construction costs to implement the project.
- Estimated duration to implement the project.
- The United States Army Corps of Engineers (USACE) National Structure Inventory (NSI) dataset to relate flood damages to building damages.
- Change in annual or periodic maintenance costs for the asset with and without the project.
- Street closures and durations associated with flooding events
- Change in maintenance costs that would be incurred after flooding events.
 - If provided by the City HDR will use provide elevation data to downtown business (i.e. doorway threshold elevations in flooding areas) to develop damage estimates.
- HDR will extract and review structure data from USACE's NSI for the project area and structures affected by flooding events. A spreadsheet model will be constructed to combine NSI data with flooding depths to estimate property damages expected for flood events.

Task 5.2 – FEMA BCA Modeling

HDR will produce a Benefit-Cost Analysis following the FEMA BCA Toolkit. The model applies methods approved by FEMA, the NRCS, and USACE. Within the model, flood inundation depths from several recurrence intervals will be used to analyze each alternative. FEMA approved depth-damage curves will be used to estimate structure, contents, and inventory damage relative to the depths of flooding above foundation by building type. The analysis will be performed for each building and flood event in the project area. Damages will be aggregated and amortized to estimate the alternative's total annual damage. Benefits will be estimated based on the difference between each proposed alternative and the no action alternative. Results of the spreadsheet model will be used in the FEMA BCA Toolkit model.

The analysis will also include modeling of travel time and increased mileage impacts from the flooding event and associated closures, and any property damages due to flooding within the project area. The effort will produce a BCA in the FEMA BCA Toolkit such that it could be submitted as part of a grant application, given all the data and information is available to complete modeling. The work will include one round of City review and HDR revisions based on City review comments.

Task 5.3 – BCA Documentation

HDR will produce documentation in support of the BCA, including a brief write-up of the project description and discussions of each benefit category, construction and maintenance costs, flooding events, and results.

Assumptions

- The City will provide requested input for the BCA, such as historical frequency of flooding events, historical maintenance costs for the asset, historical costs incurred after flooding events, and projected remaining lifespan of the asset.
- The City will provide comments on the draft BCA within 5 working days.

Deliverables

- Draft BCA results
- Final BCA results and Excel Model file
- BCA Technical Appendix (Word and PDF files)

Proposed Work Sequence and Project Schedule

Design Schedule - Amendment No.2				
	Start	Duration (work days)	End Date	Calendar Time Elapsed
Preliminary Design				
Surveying, Base Mapping, S.U.E.	3/2/2026	10	3/16/2026	14
Project Kick-off & Conceptual Design	3/11/2026	5	3/18/2026	7
Geotechnical Investigation	3/2/2026	5	3/9/2026	7
Roadway & Intersection Geometrics	3/23/2026	3	3/26/2026	3
Baseline Hydraulics & Hydrology Analysis / Drainage Impa	3/2/2026	10	3/16/2026	14
Utility Design & Coordination (BOPU)	3/23/2026	10	4/6/2026	14
60% Plans (Submittal)	4/10/2026	14	4/30/2026	20
FEMA Benefit Cost Analysis (Completed in Final Design)	6/1/2026	30	7/13/2026	42

Proposed Fee

See Attached Detailed Fee Estimate

Level of Effort and Fee Estimate
City of Cheyenne, Wyoming
Bent to 15th Street Amendment No.2 - 18th Street Reconstruction Project

	Project Team																						Subconsultant	Expenses								
Work Task	PM 7 / Engineer 11	PM 7 / Engineer 11	PM 3	Engineer 5	Engineer 1	Engineer 1	Engineer 8	Engineer 7	Engineer 6	Engineer 1	GIS 6	Economist 4	Economist 1	Economist 4	Land Surveyor 5	Land Surveyor 4	Cadd 4	Engineer 6	Engineer 2	Cadd 4	Project Accountant	Project Assistant	HDR Labor Total	Subconsultant	Mileage	Lodging	Meal Per Diem	HDR Copies	Mileage	GPS	Total Direct Expenses	Total Project
	M.Oakley	D.Hamilton	T.Mattson	B.Collamer	L. Edwards	S. Hada	T. Yerdon	J. Christopherson	B. Noble	C.Carlson	J.Hellman	D. DeFord	M. Beckner	T. Hackbart	G. Anderson	N. Fraser	J.Oakley	L. Rogers	N. Caldwell	S. Miller	E. Schramm	E. Boeve			Mile	Night	\$/Day		Mile	Hour		
Contract Billing Rate	\$ 290.00	\$ 290.00	\$ 210.00	\$ 180.00	\$ 125.00	\$ 125.00	\$ 225.00	\$ 210.00	\$ 195.00	\$ 125.00	\$ 185.00	\$ 220.00	\$ 130.00	\$ 160.00	\$ 190.00	\$ 170.00	\$ 150.00	\$ 195.00	\$ 135.00	\$ 150.00	\$ 120.00	\$ 115.00			\$ 0.725	\$ 150.00	\$ 68.00	\$ 1.00	\$ 0.88	\$ 50.00		
1.0 General Conditions	3	2	28	0	0	8	0	8	16	0	0	2	4	4	0	0	0	0	0	0	6	16	\$17,290		80	0	6	0	0	0	\$466	\$17,756
1.1 Project Management	2	2	12						8												4	4	\$5,700.00		40						\$ 29.00	
1.2 Quality Control																					2	2	\$230.00								\$ -	
1.3 Budget & Schedule Updates	1		8																		6	6	\$3,380.00								\$ -	
1.4 Progress Meetings			8			8		8	8			2	4	4								4	\$7,980.00		40		6				\$ 437.00	
2.0 Preliminary Design	0	0	18	10	38	26	0	122	22	130	9	0	0	0	80	100	80	0	0	0	0	0	\$0.00								\$ -	
2.1 Surveying, Base Mapping, S.U.E.			2		2	4			2	6	4				80	100	24						\$105,605	\$41,230	\$515	\$1,800	\$816	\$0	\$880	\$2,000	\$6,261	\$153,096
2.2 Project Kick-off & Conceptual Design			2	4	2			2		4	2												\$38,850.00	\$30,330	0	10	10		1000	40	\$ 5,060.00	
2.3 Geotechnical Investigation			2		2																		\$2,680.00								\$ -	
2.4 Roadway & Intersection Geometrics			4	4	10	6					1												\$670.00	\$10,900							\$ -	
2.5 Preliminary Plans (Submittal)			2		4	4			2	8							16						\$3,745.00								\$ -	
2.6 Baseline Hydraulics & Hydrology Analysis / Drainage Impact Study								60	4	8													\$5,210.00		60						\$ 43.50	
2.7 Utility Design & Coordination (BOPU)			2		4	4		60	10	80							16						\$14,380.00								\$ -	
2.8 60% Plans (Submittal)			4	2	16	6			4	24	2						24						\$28,370.00								\$ -	
2.9 60% Plans (Submittal)																							\$11,700.00		150						\$ 108.75	
3.0 Final Design	0	0	12	12	38	14	0	84	22	116	6	0	0	0	0	0	40	100	150	150	0	0	\$116,970	\$0	1450	2	11	330	0	0	\$2,429	\$119,399
3.1 Traffic Engineering, Lighting, Signalization			2	2	4	8																	\$2,280.00		300		6				\$ 625.50	
3.2a Detailed Underground Utility Design (Water, Sewer, BOPU)			1		4					24	2						12						\$5,880.00								\$ -	
3.2b Detailed Underground Utility Design (Storm)								24	4	40	2						12	100	150	150			\$76,580.00								\$ -	
3.3 H&H Drainage Model Updates			1					60	4	4													\$14,090.00								\$ -	
3.7 90% Plans (Submittal)			2	8	20	4			12	40	2						16						\$14,070.00		150		1				\$ 176.75	
3.9 Final Design Plans, Specifications, Quantities, Cost Est.			2	2	6	2			2	8													\$3,170.00		500		2	200			\$ 698.50	
5.0 FEMA Benefit Cost Analysis	0	0	0	0	0	0	0	0	0	0	0	16	70	60	0	0	0	0	0	0	0	0	\$22,220	\$0	300	0	6	100	0	0	\$726	\$22,946
5.1 Data Gathering												4	10	40									\$8,580.00		300		6	100			\$ 725.50	
5.2 FEMA BCA Modeling												8	40	10									\$8,560.00								\$ -	
5.3 BCA Documentation												4	20	10									\$5,080.00								\$ -	
																							\$0.00								\$ -	
Total Labor Hours per Team Member	3	2	58	22	76	48	0	214	60	246	15	18	74	64	80	100	120	100	150	150	6	16	262,085	41,230							9,882	
Total Labor Charges per Team Member	\$ 870	\$ 580	\$ 12,180	\$ 3,960	\$ 9,500	\$ 6,000	\$ -	\$ 44,940	\$ 11,700	\$ 30,750	\$ 2,775	\$ 3,960	\$ 9,620	\$ 10,240	\$ 15,200	\$ 17,000	\$ 18,000	\$ 19,500	\$ 20,250	\$ 22,500	\$ 720	\$ 1,840									Total Project	\$ 313,110